Linear and Nonlinear Relation Learning at Matched Frequency

Combined Goal 1 and Goal 2 Write-Up

factual-recall-mechanisms

2026-07-13

1 Overview

This document combines two claims tested in the same synthetic experiment. Goal 1 asks whether an infrequent, semantically supported relation can be learned in a linear format. Goal 2 asks whether a frequent relation can be fully learned while remaining nonlinear. A one-layer transformer is trained for exactly one epoch on two relations with identical frequency statistics: a semantic relation R_{sem} supported by latent attributes and a balanced random relation R_{rnd} without such support.

At F=16 occurrences per triple, R_{sem} is already learned and generalizes: trained recall is .672, held-out recall is .635, and an affine probe reaches 0.990 ± 0.007 accuracy. The frequency-matched random relation is barely learned and its affine probe remains at chance. This is the Goal 1 infrequent/linear result. At F=64, both relations reach perfect recall on all trained triples, but the held-out affine probe is 0.891 ± 0.056 for R_{sem} and 0.016 ± 0.022 for R_{rnd}. This is the Goal 2 frequent/nonlinear result.

2 Shared experimental setup

2.1 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; and 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries $(f_{i \bmod 32}, g_{\lfloor i/32 \rfloor})$. The 192 subjects therefore realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32}, g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. For seed 19, `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5` while R_{rnd}(`a_37`) = `b_27`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are trained and 2 are held out: 128 trained subjects and 64 held-out subjects. The random mapping is balanced within both groups. Under both relations, every trained subject occurs in exactly one triple, every object is the target of exactly 4 trained and 2 held-out subjects, and both relations contribute 128 training triples repeated the same number of times F. Held-out subjects appear in attribute sentences but never in relation triples. These invariants are asserted by `tests/test_linear_shift.py`.

2.2 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$, $F \in \{1, 4, 16, 64\}$
Training	exactly one epoch: one pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64), batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, LR 10^{-3} after 5 warmup steps, weight decay 0,
	gradient clip 1.0
Seeds	19, 31, 47

At F=1, every relation triple occurs exactly once in the model's complete training history. At F=64, each triple occurs 64 times, but training is still a single pass over one corpus.

2.3 Metric and learning check

The single analysis metric operationalizes the claim that a relation is an affine transformation. Ordinary least-squares regression fits

$$\hat{o} = \hat{W}s + \hat{b}$$

from the subject input embedding $s_i \in \mathbb{R}^{64}$ to the target object's unembedding row $o_{R(i)} \in \mathbb{R}^{64}$ using the 128 trained pairs. For each of the 64 held-out subjects, the predicted vector is matched by cosine similarity against all 32 object rows. Accuracy is the fraction for which the nearest row is the unique correct object. Chance is $1/32 \approx 0.03$.

Greedy next-token recall on prompts `x a_i r_r` is recorded as a learning check. It distinguishes a relation whose representations admit an affine map from a relation the model can actually recall.

3 Goal 1: An infrequent relation is learned as linear

3.1 Claim

At low fact frequency, F=16 occurrences per triple in one epoch, R_{sem} is genuinely learned: recall is .672 on trained triples and .635 on held-out subjects never seen in a relation triple. What is learned is linear: the held-out affine probe reaches 0.990 ± 0.007. The frequency-matched random control is barely learned (trained recall .167) and its probe remains at chance.

		probe accuracy (held-out)		recall (trained)		recall (held-out)	
	F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
	1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
	4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
	16	$\mathbf{0.990 \pm .007}$	$.036 \pm .007$	**.672**	.167	**.635**	.031
control: no triples		$1.000 \pm .000$	$.021 \pm .015$	–	–	.000	.000

Table 1: Goal 1 low-frequency cells, mean ± sd over seeds 19/31/47 at A=128. The headline cell is F=16: R_{sem} is learned and linear, while the frequency-matched R_{rnd} is barely learned and nonlinear.

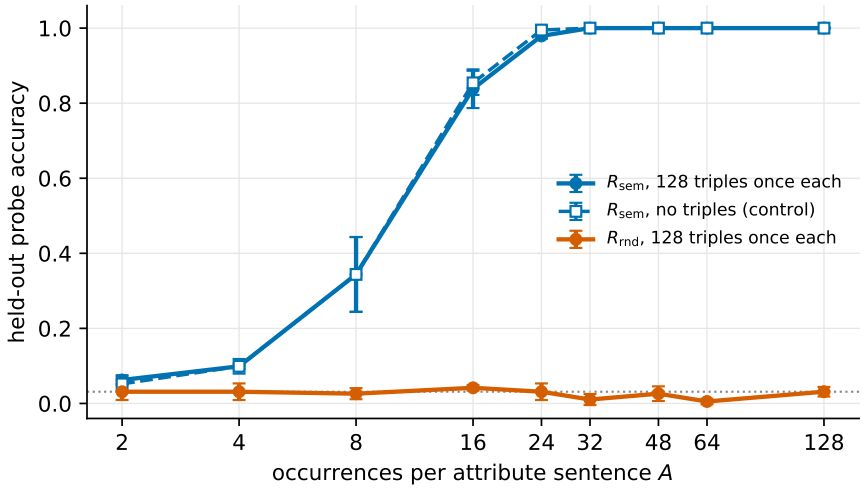


Figure 1: At F=1, probe accuracy as a function of attribute-sentence occurrences A. The semantic transition from chance to ceiling is sharp and is identical with 128 once-occurring triples (solid) and no triples (dashed control). The random relation remains at chance.

3.2 Results

Finding 1: at low frequency, what is learned is an affine map. With each fact occurring only 16 times, R_{sem} is learned on trained subjects, generalizes to held-out subjects, and supports an almost perfect held-out affine map. The frequency-matched random control is barely learned and remains at chance under the probe. The semantic relation is learned earlier because it can use the linear attribute structure.

Finding 2: the linear format is present at single occurrence. At F=1, the probe is already 1.000 for R_{sem} and at chance for R_{rnd}. This is a representational statement: the model does not yet reproduce individual triples behaviorally at F=1 (recall approximately .03).

Finding 3: the transition is driven by attribute knowledge. Sweeping A at F=1 produces the probe sequence $0.05 \to 0.10 \to 0.34 \to 0.85 \to 1.00$ for $A = 2, 4, 8, 16, 24$. The zero-triple control follows the same curve, showing that the affine subject-to-object geometry is carried by attribute training rather than triple repetition.

3.3 Goal 1 scope

The experiment shows an infrequent relation learned in a linear, generalizing format and a sharp transition into the linear regime. It does not show that single-occurrence relation triples cause that transition: the no-triple control attributes it to scaffold exposure. Earlier single-exposure phase-shift results using growing probe sets and attribute-initialized embeddings are superseded.

4 Goal 2: A frequent relation is learned as nonlinear

4.1 Claim

At high fact frequency, F=64, both relations are fully learned: recall on all 128 trained triples is 1.000 for both relations in all three seeds. Nevertheless, the held-out affine probe reaches 0.891 ± 0.056 for R_{sem} and only 0.016 ± 0.022 for R_{rnd}, equal to chance. The frequent random relation is thus stored without an affine map over the evaluated representations and does not generalize to held-out subjects.

4.2 Results

	probe accuracy (held-out)		recall (trained)		recall (held-out)	
F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
16	$0.990 \pm .007$	$.036 \pm .007$	.672	.167	.635	.031
64	$0.891 \pm .056$	$\mathbf{.016 \pm .022}$	**1.000**	**1.000**	.995	**.031**

Table 2: Goal 2 frequency sweep, mean ± sd over seeds 19/31/47 at A=128. At F=64, both relations are fully recalled on trained triples, but only R_{sem} admits a held-out affine map.

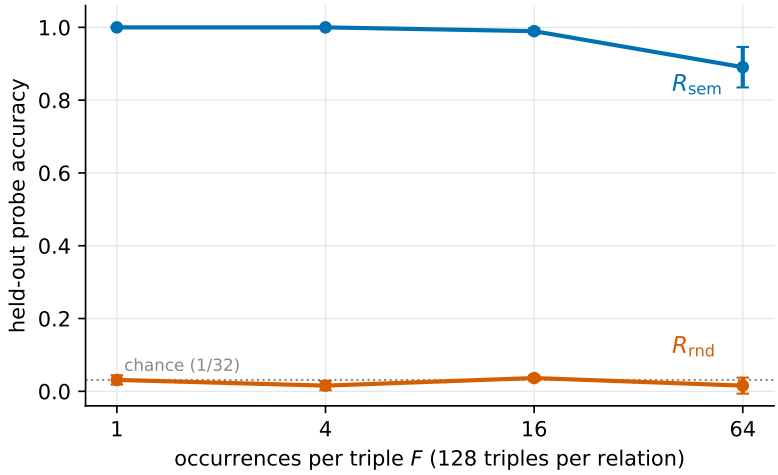


Figure 2: Held-out probe accuracy versus occurrences per triple for the two frequency-matched relations. The random relation stays at chance while the semantic relation remains high, with mild erosion at F=64.

Finding 1: a fully learned frequent relation without held-out affine structure. At 64 occurrences per triple, R_{rnd} reaches perfect recall of every trained triple while its held-out probe and held-out behavioral recall remain at chance. It is therefore a non-generalizing memorization under the stated affine metric.

Finding 2: the contrast is due to semantics, not frequency. The semantic relation is trained in the same model with exactly the same frequency, is equally well recalled, and retains high affine-probe and held-out recall. Raising R_{rnd} from 1 to 64 occurrences changes whether it is recalled but never lifts its held-out affine score above chance.

Finding 3: high frequency mildly erodes semantic linearity. The R_{sem} probe declines from 1.000 at $F \leq 4$ to 0.891 ± 0.056 at F=64, consistent with high-frequency memorization competing with the linear solution while leaving it far above the random relation.

4.3 Goal 2 scope

The precise claim is not that frequency always causes nonlinearity. At matched high frequency, the attribute-supported relation remains largely linear while the unsupported random relation is fully recalled without held-out affine transfer. Earlier high-frequency experiments using structured embedding initialization, targeted-row updates, or top-k metrics are superseded by this end-to-end setup.

5 Joint conclusion

The two goals establish complementary regimes in one frequency-controlled experiment. Semantic structure permits an affine, generalizing solution before the random control is well learned. With enough repetition, the random relation can be perfectly recalled on its training facts without

acquiring held-out affine transfer. Frequency controls whether unsupported facts are memorized; semantic support controls whether the learned relation aligns with a transferable affine geometry.

6 Reproduction

Implementation: `experiments/factual_recall/linear_shift.py`.

Design invariants: `experiments/tests/test_linear_shift.py`.

Run the following commands from `experiments/`:

```
# Tables and Goal 2 frequency figure; run separately for seeds 19, 31, and 47
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1,4,16,64 --attr-repeats 128 \
  --device cuda --output-dir factual_recall/results/final_category_seed19

# Goal 1 transition figure; repeat for each seed and A in
# {2,4,8,16,24,32,48,64,128}
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1 --attr-repeats 8 \
  --device cuda \
  --output-dir factual_recall/results/final_category_arsweep_seed19_ar8

uv run pytest tests/test_linear_shift.py -q
```

Raw metrics are in `factual_recall/results/final_category_*/linear_shift_metrics.json`. The source claims remain available separately in `goal1_infrequent_relation_linear.tex` and `goal2_frequent_relation_nonlinear.tex`.